

Q1. How does Computer Forensics help in Law Enforcement?

Computer forensics helps law enforcement by finding and analyzing digital evidence from computers and other devices.

Main points:

1. **Recovers deleted files** – Finds deleted documents, photos, and other files.
2. **Finds hidden data** – Searches hidden files and unallocated disk space.
3. **Traces computer activity** – Finds information left behind by the operating system.
4. **Checks file history** – Finds when files were created, modified, accessed, or deleted.
5. **Finds emails** – Can search and recover email-related evidence.
6. **Provides evidence** – Helps investigators collect useful digital evidence for criminal cases.

Below are **Q2–Q10 answers based on your uploaded PDF**, written in **easy words, short, and exam-oriented for 5 marks**. I have kept the same terminology used in the PDF.

Q2. Explain the application of Digital Forensics in the Military. [5 Marks]

Digital forensics helps the military to investigate **cyber attacks and malicious activities**.

Applications:

1. **Find digital evidence** – Quickly discovers evidence related to cyber attacks.
2. **Find attacker identity** – Helps identify who performed the attack.
3. **Find attacker location** – Helps determine the possible location of the attacker.
4. **Understand attacker's intention** – Helps understand the motive and purpose of the attack.
5. **Assess damage** – Helps estimate the impact of malicious activity on the victim/system.
6. **Track hidden activity** – Helps investigate data that has been hidden, destroyed, or modified.

In short: Military digital forensics helps **find evidence, identify attackers, understand their intention, and assess cyber-attack impact**.

Q3. Explain the various business-oriented digital forensic techniques. [5 Marks]

The main business-oriented digital forensic techniques are:

1. **Remote Monitoring** – Monitors computer activities remotely.
2. **Trackable Electronic Documents** – Helps track unauthorized access, download, or viewing of documents.
3. **Theft Recovery Software** – Helps locate lost or stolen computers.
4. **Network Monitoring** – Monitors remote PCs and network activities.

5. **Tracking Stolen Files** – Helps find the location of stolen electronic files.
 6. **Honeypot Operations** – Helps identify unauthorized users or attackers.
 7. **Protection from Hackers and Viruses** – Helps protect company systems and data.
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Q4. What are the various data recovery solutions? [5 Marks]

Data recovery means recovering lost or damaged data and making it usable again.

Various solutions are:

1. **Backup** – Keep copies of important data for recovery.
 2. **Image Copy** – Create copies/snapshots of data.
 3. **Change Accumulation** – Store changes made to data for recovery.
 4. **Recovery** – Recover databases and data to a required point in time.
 5. **Recovery Manager** – Automates and manages the recovery process.
 6. **Index Rebuild** – Rebuilds damaged or lost database indexes.
 7. **Recovery Advisor** – Checks whether databases are properly backed up.
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Q5. What is the significance of Data Recovery and Backup? [5 Marks]

Data Recovery: It is the process of recovering lost, deleted, or damaged data.

Importance of Backup and Recovery:

1. **Prevents data loss** – Keeps a safe copy of important data.
2. **Recovers deleted/lost data** – Helps restore required files.
3. **Protects against system failure** – Helps after hardware or software failures.
4. **Reduces downtime** – Helps restore systems quickly.
5. **Protects against disasters** – Helps recover data after disasters.
6. **Handles user errors** – Allows recovery from accidental mistakes.

In short: Backup keeps a safe copy, while recovery helps **restore data when a problem occurs**.

Q6. What is Computer Forensics? What are its broad steps? [5 Marks]

Computer Forensics:

Computer forensics is the process of **collecting, preserving, analyzing, and presenting computer-related evidence**.

Broad Steps:

1. **Protect the system** – Prevent alteration or damage to evidence.
 2. **Discover files** – Find normal, deleted, hidden, protected, and encrypted files.
 3. **Recover deleted files** – Restore deleted files.
 4. **Reveal hidden data** – Examine hidden and temporary files.
 5. **Analyze data** – Examine relevant data from the computer.
 6. **Prepare report** – Document findings and relevant files.
 7. **Expert testimony** – Explain findings when required.
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Q7. How can employers ensure security of their company data through deploying digital forensics?
[5 Marks]

Employers can use digital forensics to protect important company data.

1. **Create exact duplicate** – Make an exact copy of an employee's computer data when required.
2. **Protect original data** – Work on the duplicate so the original remains unchanged.
3. **Recover deleted data** – Recover damaged or deleted company information.
4. **Find data theft** – Detect removal of confidential/proprietary information.
5. **Investigate employee activity** – Check computer activity and identify misuse.
6. **Protect against false claims** – Digital evidence can help prove what actually happened.

In short: Digital forensics helps employers **protect, recover, monitor, and investigate company data**.
